

Money, Banks, and the Bank of Canada

ECON 101 - Macroeconomics

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- 10.1 What Is Money, and Why Do We Need It?
- 10.2 How Is Money Measured in Canada Today?
- 10.3 How Do Banks Create Money?
- 10.5 The Quantity Theory of Money

10.1 Learning Objective

Goal

Define money and discuss the four functions of money.

- Understand what economists mean by **money**.
- See how money solves problems with barter.
- Learn the **four functions** of money.
- Discuss what can and cannot serve as money.

What Is Money?

- **Money** is **any asset** that people are generally willing to accept:
 - in exchange for goods and services, or
 - in payment of debts.
- **Asset:** anything of value owned by a person or a firm.
- Money is one of the most important inventions in economic history because:
 - it makes trade easier,
 - it allows specialization,
 - it supports the development of complex economies.

Life Without Money: Barter

- Imagine living before money existed.
- All trade would be **barter**:
 - trading goods and services directly for goods and services.
- Barter requires a **double coincidence of wants**:
 - you must have what I want,
 - and I must have what you want,
 - at the same time and place.
- Barter works in small, simple economies, but breaks down as:
 - the number of goods and services grows,
 - and people become more specialized.

From Barter to Commodity Money

- To avoid the problems of barter, societies adopted **commodity money**.
- **Commodity money:**
 - goods used as money that also have value independent of their use as money.
- Examples:
 - Precious metals such as **gold** and **silver**,
 - Beaver pelts among early settlers and some Indigenous peoples in Canada,
 - Cigarettes in prisons and prisoner-of-war camps.
- Commodity money works as long as people:
 - value the commodity itself,
 - and agree to use it in trade.

The Four Functions of Money

- Money fulfills **four primary functions**:
 1. **Medium of exchange**
 2. **Unit of account**
 3. **Store of value**
 4. **Standard of deferred payment**

Function 1: Medium of Exchange

- Money is a **medium of exchange** if:
 - it is acceptable to a wide variety of parties
 - as payment for goods and services.
- Instead of barter:
 - you sell your labour or goods $\Rightarrow$ receive money,
 - then use money to buy what you want.
- This greatly reduces the time and effort needed to find trading partners.

Function 2: Unit of Account

- Money is a **unit of account**:
 - it provides a standard way of measuring the value of goods and services.
- In Canada, prices are quoted in \$ **Canadian dollars**.
- Without a unit of account:
 - every good would have to be priced in terms of every other good.
 - In a large economy, this would be extremely confusing.

Function 3: Store of Value

- Money is a **store of value**:
 - it allows people to transfer purchasing power from today to the future.
- Other assets also store value:
 - stocks, bonds, houses, land, art, etc.
- Money is special because it is:
 - highly **liquid** (easily converted into goods and services),
 - relatively stable in value in low-inflation economies.
- Inflation reduces the effectiveness of money as a store of value.

Function 4: Standard of Deferred Payment

- Money is a **standard of deferred payment**:
 - it is widely accepted for settling debts that are paid back in the future.
- Loans and credit contracts are typically denominated in money terms.
- This works well when:
 - the value of money is reasonably **predictable** over time.

What Can Serve as Money?

- For an asset to serve as a widely accepted medium of exchange, it should:
 1. Be **acceptable** to most people.
 2. Have **standardized quality** so any two units are alike.
 3. Be **durable** so it does not wear out quickly.
 4. Be **valuable relative to its weight**, so it is easy to transport.
 5. Be **divisible** enough to make both small and large transactions.
- Historical and modern monies (gold coins, bank notes, electronic deposits) satisfy these conditions to different degrees.

Commodity Money: Examples

- **Commodity money** has value independent of its use as money.
- Examples:
 - Gold and silver coins.
 - Beaver pelts in early Canadian trade.
 - Cigarettes used in prisons or war camps.
- Advantage:
 - intrinsic value supports trust in the money.
- Disadvantages:
 - can be costly to store and transport,
 - supply may be hard to control.

- In modern economies, money is typically **paper currency** issued by a **central bank**.
- In Canada, the central bank is the **Bank of Canada**.
- Modern currencies are **not** convertible into gold or any commodity.
- **Fiat money:**
 - money, such as paper currency, that is authorized by a central bank or government,
 - and does not have to be exchanged for a commodity like gold.

Fiat Money (2 of 2)

- Fiat money is more **flexible**:
 - central banks can adjust the money supply to respond to shocks.
- But it also has a potential problem:
 - it is valuable only as long as households and firms **have confidence** in it.
- If people lose confidence that paper dollars will hold their value:
 - they will not accept them in exchange,
 - and fiat money will cease to be useful.

10.2 Learning Objective

Goal

Discuss the definitions of the money supply used in Canada today.

- Different assets vary in how **liquid** they are.
- The Bank of Canada uses several definitions of the money supply:
 - from narrow measures (very liquid) to broad measures (include less liquid assets).

Figure 10.1 – Components of the Money Supply (1 of 3)

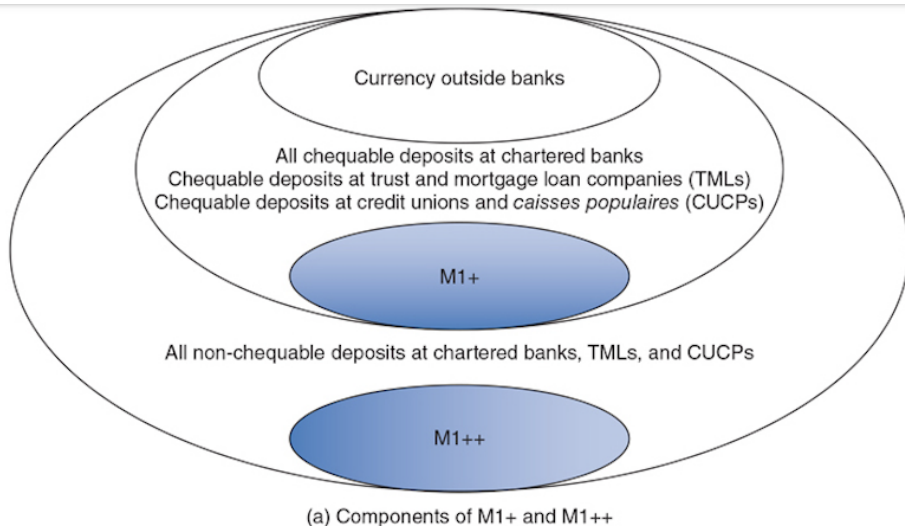
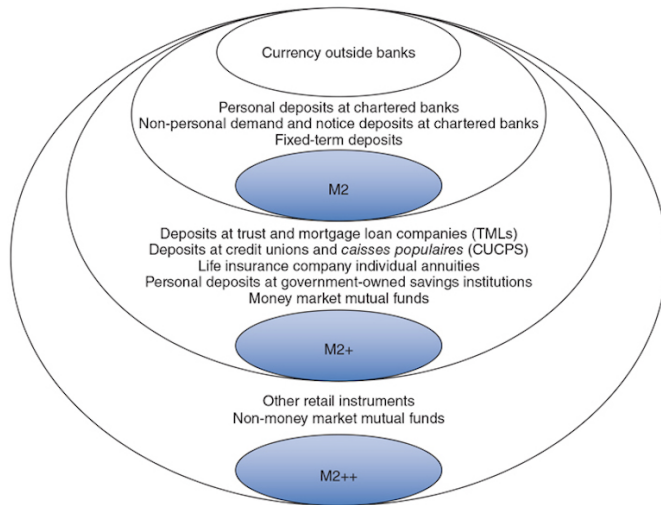
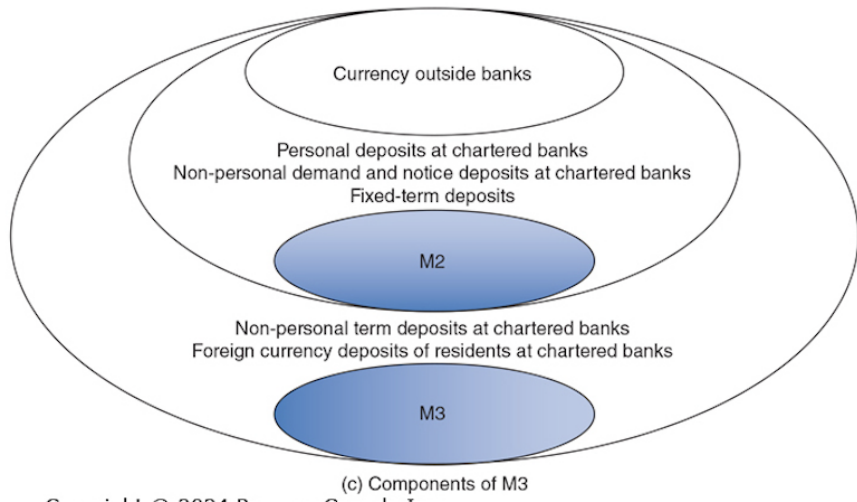


Figure 10.1 – Components of the Money Supply (2 of 3)



(b) Components of M2, M2+, and M2++

Figure 10.1 – Components of the Money Supply (3 of 3)



The Bank of Canada's Measures of Money

- The Bank of Canada uses six main measures:
 - **M1+**, **M1++**, **M2**, **M2+**, **M2++**, **M3**.
- **M1+**: narrowest definition.
- **M2++**: broadest definition.
- Each broader measure includes all the assets in the narrower measures, plus some extra, less liquid assets.

Definition of M1+ and M1++

- **M1+:**
 - Currency in circulation (bank notes and coins outside banks).
 - **Chequable deposits** at:
 - banks,
 - trust and mortgage companies,
 - credit unions and *caisses populaires*.
- **M1++:**
 - Includes all of M1+,
 - plus **non-chequable deposits**.

Definition of M2 and M2+

- **M2** includes:
 - Currency,
 - Personal deposits at banks,
 - Non-personal demand and notice deposits at banks,
 - Fixed-term deposits at banks.
- **M2+**: includes M2 plus:
 - Deposits at trust and mortgage loan companies (TMLs),
 - Deposits at credit unions and *caisses populaires* (CUCPs),
 - Life insurance company individual annuities,
 - Government-owned savings institutions,
 - Money market mutual funds.

Definition of M2++ and M3

- **M2++** (broadest measure):
 - Includes M2+ plus:
 - Canada Savings Bonds,
 - Non-money market mutual funds.
- **M3** includes:
 - M2,
 - Non-personal term deposits at banks,
 - Foreign-currency deposits of Canadians at banks.

Which Money Supply Measure to Use?

- Any of the main measures might be relevant, depending on the question.
- For most macroeconomic analysis, we focus on money as a **medium of exchange**.
- That suggests concentrating on **M1+**:
 - currency in circulation,
 - chequable deposits.
- In our discussion:
 - we treat both currency and chequing/non-chequing account balances as “money”,
 - but not the broader set of financial assets.

10.3 Learning Objective

Goal

Explain how banks create money.

- Banks are central to the money creation process.
- More money is held in chequing accounts than exists as physical currency.
- Banks are profit-making private firms; they:
 - accept deposits,
 - make loans,
 - and thereby **create money**.

Figure 10.2 – Balance Sheet for Royal Bank

Asset (in billions)		Liabilities and Shareholders' Equity (in billions)	
Reserves	\$199	Deposits	\$1082
Loans	950	Borrowing	24
Securities	282	Other liabilities	531
Other assets	302		
		Total liabilities	\$1637
		Shareholders' equity	96
Total assets	\$1733	Total liabilities and shareholders' equity	\$1733

Bank Balance Sheets: Assets and Liabilities

- On a **balance sheet**:
 - Assets are listed on the left,
 - Liabilities and stockholders' equity (**net worth**) on the right.
- **Assets** for a bank include:
 - Reserves (cash in vaults and deposits at the Bank of Canada),
 - Loans,
 - Securities and other investments.
- **Liabilities** include:
 - Deposits (chequing, savings, term deposits),
 - Borrowings.
- The two sides must always sum to the same total.

Reserves and Desired Reserve Ratio

- **Reserves:**
 - deposits that a bank keeps as cash in its vault or on deposit with the Bank of Canada.
- Banks in Canada are **not** required to hold a specific level of reserves by law.
- They choose to keep a fraction of deposits on hand as reserves:
 - often around 5%.
- **Desired reserve ratio (r_d):**
 - the fraction of deposits banks *want* to keep as reserves.
- Banks may adjust r_d depending on:
 - the state of the economy,
 - and financial market conditions.

T-Accounts: A Simple Tool

- A **T-account** is a simplified balance sheet.
- It shows only how a transaction **changes** assets and liabilities.
- Example: you deposit \$1,000 in currency at Bank of Montreal (BMO).
- BMO's T-account change:

Assets	Liabilities
Reserves +\$1,000	Deposits +\$1,000

- The currency component of the money supply falls by \$1,000, but chequing deposits rise by \$1,000, so total money supply is unchanged **so far**.

From Deposits to Loans: Creating New Money

- BMO holds some of the \$1,000 as reserves and lends out the rest.
- Suppose it keeps 10% (\$100) as reserves and lends \$900.
- T-account changes for BMO after the loan:

Assets	Liabilities
Reserves +\$1,000	Deposits +\$1,000
Loans +\$900	Deposits +\$900

- The \$900 loan is typically deposited at another bank, so:
 - the banking system now has:
 - \$1,000 in new reserves,
 - \$1,900 in deposits.

Multiple Rounds of Money Creation

- The process continues across banks:
 - each bank keeps 10% of new deposits as reserves,
 - and lends out the remaining 90%.
- First deposit: \$1,000 reserves.
- First loan: \$900 new deposit in the system.
- Second round:
 - 10% of \$900 (i.e., \$90) kept as reserves,
 - \$810 loaned out and deposited.
- Third round:
 - 10% of \$810 kept as reserves,
 - \$729 loaned out, etc.

When Will It End?

- Each round of lending creates smaller and smaller new deposits.
- With a 10% desired reserve ratio ($r_d = 0.10$):
 - the original \$1,000 in currency becomes the 10% reserves for all the new deposits.
- So total checking deposits created will be:

$$\text{Total deposits} = \frac{1}{r_d} \times \text{new reserves} = \frac{1}{0.10} \times \$1,000 = \$10,000.$$

- The banking system has created \$9,000 of new money (deposits) on top of the original \$1,000.

Simple Deposit Multiplier

- The total increase in deposits from an initial increase in reserves can be written as:

$$\Delta \text{Deposits} = \Delta \text{Reserves} + (1 - r_d) \Delta \text{Reserves} + (1 - r_d)^2 \Delta \text{Reserves} + \dots$$

- This is a geometric series whose sum is:

$$\Delta \text{Deposits} = \frac{1}{r_d} \times \Delta \text{Reserves}.$$

- The **simple deposit multiplier** is:

$$\text{Simple deposit multiplier} = \frac{1}{r_d}.$$

Simple Deposit Multiplier: Numerical Examples

- If $r_d = 0.10$ (10% desired reserve ratio):

$$\text{Simple deposit multiplier} = \frac{1}{0.10} = 10.$$

- If $r_d = 0.20$ (20% desired reserve ratio):

$$\text{Simple deposit multiplier} = \frac{1}{0.20} = 5.$$

- Example:
 - New deposits of \$100,000, with $r_d = 0.10$,
 - can support total deposits of:

$$\frac{1}{0.10} \times \$100,000 = \$1,000,000.$$

Simple vs. Real-World Deposit Multiplier

- The simple deposit multiplier is an **upper bound**.
- In practice, the real-world deposit multiplier is **smaller** because:
 1. Banks may choose to hold **excess reserves** (more than desired).
 2. Some of the money loaned out is held as **currency** rather than redeposited in banks.
- During financial crises, banks may be very cautious:
 - the real-world multiplier can fall close to 1.

Conclusions on Banks and the Money Supply

- In general, the real-world deposit multiplier is greater than 1.
- Therefore:
 - When banks gain reserves, they make new loans, and the money supply **expands**.
 - When banks lose reserves, they reduce loans, and the money supply **contracts**.
- This tight connection between reserves, lending, and deposits means:
 - central bank actions that affect bank reserves also affect the money supply.

10.5 Learning Objective

Goal

Explain the quantity theory of money and use it to explain how high rates of inflation occur.

- Link money growth to inflation in the long run.
- Understand the **quantity equation**.
- See how hyperinflation arises when money supply growth is excessive.

Historical Motivation: Gold and Silver Flows

- Beginning in the 16th century, Spain shipped gold and silver from Mexico and Peru to Europe.
- These precious metals were minted into coins, increasing the money supply.
- Over these years, prices in Europe rose steadily.
- Observers started to connect:
 - increases in the quantity of money,
 - with increases in the overall price level.

The Quantity Equation

- In the early 20th century, Irving Fisher formalized this relationship in the **quantity equation**:

$$M \times V = P \times Y,$$

where:

- M = money supply,
 - V = velocity of money,
 - P = price level,
 - Y = real output (real GDP).
- **Velocity of money:**
 - the average number of times per year each dollar in the money supply is used to purchase goods and services included in GDP.

Calculating the Velocity of Money

- Rearranging the quantity equation:

$$V = \frac{P \times Y}{M}.$$

- If we measure:
 - M with M1+,
 - P with the GDP deflator,
 - Y with real GDP,
- then we can compute V for any year.
- In practice, V fluctuates from year to year, but it is often fairly stable over longer periods in some economies.

- The **quantity theory of money** assumes that the velocity of money is approximately constant in the long run.
- Under this assumption:

$$M \times V = P \times Y,$$

- with V treated as constant.
- Changes in M and Y must then translate into changes in P .

From Levels to Growth Rates

- When variables are multiplied in an equation, we can write an equivalent relationship in terms of their **growth rates**.
- Let:
 - g_M = growth rate of money,
 - g_V = growth rate of velocity,
 - π = inflation rate (growth rate of the price level P),
 - g_Y = growth rate of real GDP.
- Then the quantity equation implies:

$$g_M + g_V = \pi + g_Y.$$

Inflation According to the Quantity Theory

- If we assume that V is constant in the long run:

$$g_V = 0,$$

- then the growth-rate form simplifies to:

$$g_M = \pi + g_Y.$$

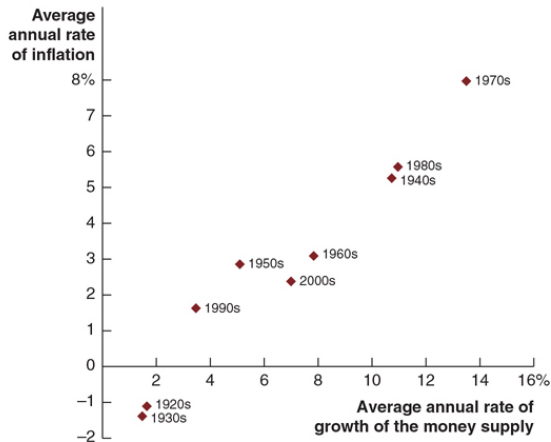
- Solving for inflation:

$$\pi = g_M - g_Y.$$

- Predictions:

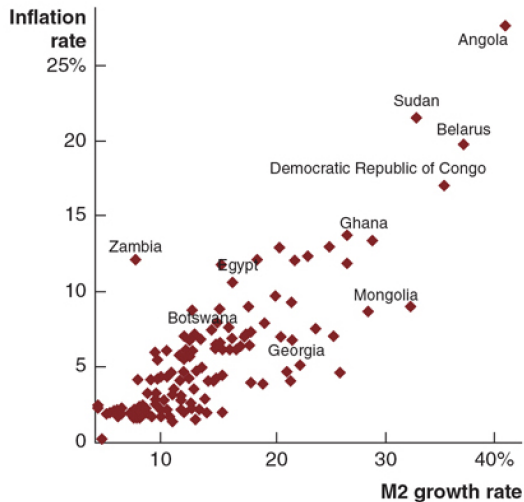
- If $g_M > g_Y$: $\pi > 0$ (inflation).
- If $g_M < g_Y$: $\pi < 0$ (deflation).
- If $g_M = g_Y$: $\pi = 0$ (stable price level).

Figure 10.7 – Money Growth and Inflation (1 of 2)



(a) Inflation and money supply growth in Canada, 1920s–2010s

Figure 10.7 – Money Growth and Inflation (2 of 2)



Money Growth and Inflation: Interpretation

- Real GDP growth (g_Y) tends to be relatively stable over long periods.
- The quantity theory therefore predicts:
 - a broadly positive relationship between money growth and inflation.
- Empirically:
 - countries with higher average money growth have higher average inflation,
 - especially when looking at long time spans or very high inflation episodes.

High Inflation and Hyperinflation

- **Hyperinflation:**
 - extremely high inflation, often defined as **more than 50% per month**.
- Hyperinflation typically occurs when:
 - central banks increase the money supply far faster than real GDP,
 - often because governments are financing very large budget deficits by printing money.
- Example:
 - Zimbabwe experienced hyperinflation so severe that people stopped using the national currency.
- Hyperinflation is usually associated with:
 - very poor economic performance,
 - deep recessions or even economic collapse.

Quantity Theory: Main Takeaway

- The quantity theory of money is a **long-run** theory.
- It emphasizes that:
 - **persistent inflation** is ultimately a monetary phenomenon.
 - In the long run, sustained high inflation requires:

$$g_M \gg g_Y.$$

- Central banks that keep money growth close to real GDP growth:
 - can maintain low and stable inflation over time.

Summary

- Money is any asset widely accepted in exchange for goods, services, or debt payments.
- Money performs four key functions:
 - medium of exchange,
 - unit of account,
 - store of value,
 - standard of deferred payment.
- Canada's money supply has several measures; we emphasize M1+ for its role as a medium of exchange.
- Banks create money by accepting deposits and making loans, subject to their desired reserve ratio.
- The simple deposit multiplier equals $1/r_d$, but the real-world multiplier is smaller.
- The quantity theory of money links long-run inflation to the growth rate of the money supply relative to real GDP.

Thank you!